Introduction to Cryptography

Lecture 35: Introduction to Public-Key Cryptography

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Learning Outcomes

- Explain why public-key cryptography was proposed.
- Identify the six ingredients of a public-key cryptosystem.
- Distinguish confidentiality from authentication.
- Compare symmetric and asymmetric cryptography.

Two Problems That Motivated Public-Key Cryptography

- Symmetric encryption needs a pre-shared secret or a trusted Key Distribution Center.
- Digital communication also needs a way to authenticate a signer.
- Diffie and Hellman publicly introduced public-key cryptography in 1976.

Symmetric vs. Asymmetric Keys

Symmetric	Asymmetric
Same secret key on both sides	Public/private key pair
Secret key must be shared securely	Public key may be published
Efficient for bulk data	Often used for key establishment and signatures

Six Ingredients

- Plaintext
- Encryption algorithm
- Public key
- Private key
- Ciphertext
- Decryption algorithm

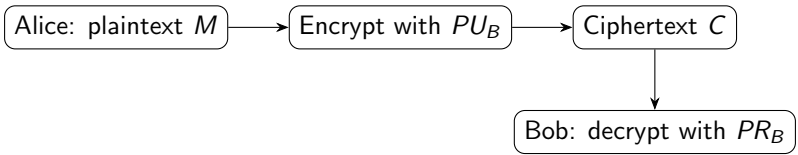
Key Pair and Notation

$$PU_B = \text{Bob's public key}, \quad PR_B = \text{Bob's private key}$$

$$C = E(PU_B, M), \qquad M = D(PR_B, C)$$

Public key is distributed; private key must remain confidential.

Confidentiality: Alice Sends to Bob



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Public-Key Distribution

- Bob creates a mathematically related key pair.
- Bob publishes the public key through a trusted mechanism.
- Alice verifies that the public key really belongs to Bob.
- Alice encrypts to Bob; only Bob holds the corresponding private key.
- Warning: an unauthenticated public key can be replaced by an attacker.

Authentication and Digital Signatures

- A signer uses a private key to generate a signature.
- A verifier uses the corresponding public key to check it.
- Modern signatures operate on a digest with a defined signature scheme.
- Signing does not conceal the message.

Encryption Is Not Signing

Confidentiality

Encrypt to the recipient using the recipient's public key.

Authenticity

Sign using the sender's private key; verify using the sender's public key.

Do not use raw RSA private-key exponentiation as a practical signature protocol.

Hybrid Cryptography

1. Generate a random symmetric session key.
2. Protect or establish that key using public-key cryptography.
3. Encrypt bulk data with an authenticated symmetric cipher.

Reason

Public-key operations are typically more computationally expensive.

Threats and Assumptions

- Protect private keys against theft.
- Authenticate public keys to prevent substitution.
- Choose secure parameters and padding.
- Account for side-channel and implementation attacks.

Quick Check

1. Which key does Alice use to encrypt confidentially for Bob?
2. Which key does Bob use to decrypt?
3. Does a valid digital signature hide message contents?
4. Why use a hybrid cryptosystem?

Summary

$$\boxed{C = E(PU_B, M), \quad M = D(PR_B, C)}$$

Public-key cryptography helps with confidentiality, signatures, and key establishment.

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